Secure Application Design

Apertura: martes, 17 de marzo de 2026, 08:30
Cierre: lunes, 23 de marzo de 2026, 23:59

Enterprise Architecture Workshop: Secure Application Design

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In this workshop, we will design and deploy a secure, scalable application using AWS infrastructure with a focus on best practices for security. Our architecture will feature two primary components:

1. **Server 1: Apache Server**
   The Apache server will be responsible for serving an asynchronous HTML+JavaScript client over a secure connection using TLS. Client-side code will be delivered through encrypted channels, ensuring data integrity and confidentiality during download.
2. **Server 2: Spring Framework**
   The Spring server will handle backend services, offering RESTful API endpoints. These services will also be protected using TLS, ensuring secure communication between the client and the backend.

Key Security Features:

- **TLS Encryption:** Secure transmission of data using TLS certificates generated through Let's Encrypt, ensuring confidentiality and integrity.
- **Asynchronous Client:** Our HTML+JavaScript client will leverage async techniques to optimize performance while maintaining secure communication.
- **Login Security:** We will implement login authentication, with passwords securely stored as hashes.
- **AWS Deployment:** All services will be deployed and managed on AWS, leveraging its secure, reliable infrastructure.

This workshop will guide participants through the process of integrating these security measures, configuring multi-server deployments, and utilizing modern encryption techniques to protect user data.

Hint: https://docs.aws.amazon.com/linux/al2023/ug/ec2-lamp-amazon-linux-2023.html

Hint 2: Follow the attached turorial TallerAllSecureAppSpring

Rubric

Class Work (50%)

- **Participation & Collaboration (20%)**
  - Actively engages in discussions and group activities.
  - Collaborates effectively with peers during hands-on exercises.
  - Contributes ideas and solutions during architecture design sessions.
- **Hands-on Lab Performance (30%)**
  - Successfully deploys the application on AWS using the provided instructions.
  - Demonstrates proper configuration of Apache and Spring on separate servers.
  - Configures TLS for both client download from Apache and REST service requests from Spring.
  - Implements login security with hashed password storage.
  - Generates and installs Let's Encrypt certificates for both servers.
  - Submits all relevant code to a GitHub repository, including clear documentation.

Homework (50%)

- **Application Architecture Design (25%)**
  - Provides a detailed design document for the application architecture.
  - Correctly outlines the relationship between Apache, Spring, and the async HTML+JS client.
  - Demonstrates understanding of secure deployment strategies on AWS.
- **Security Implementation (15%)**
  - Submits a fully working application that adheres to all security requirements.
  - Configures and demonstrates the use of TLS for secure connections between the client, Apache, and Spring.
  - Implements login functionality with passwords securely stored as hashes.
  - Utilizes Let's Encrypt for certificate management.
- **Final Deliverables (10%)**
  - GitHub repository containing all source code, a README with instructions for deployment, architecture overview, and screenshots from testing.
  - A video demonstrating the application deployment and explaining the security features.

Editar entrega Borrar entrega

Estado de la entrega

Estado de la entrega	Enviado para calificar
Estado de la calificación	Sin calificar
Tiempo restante	La tarea fue enviada 5 horas 44 minutos antes de la fecha límite
Última modificación	lunes, 23 de marzo de 2026, 18:14
Texto en línea	+ https://github.com/JAPV-X2612/AREP-laboratory-6-secure-application-design
Comentarios de la entrega	Comentarios (0)

Enlaces de interés

Ministerio de Educación Nacional

Colombia Aprende

Red Latinoamericana de Portales Educativos

Red Universitarias Metropolitana de Bogotá

Biblioteca

Investigación e innovación

Enlace - Académico

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